

3.8 Maxima and Minima of Functions of Two Variables

Definition:

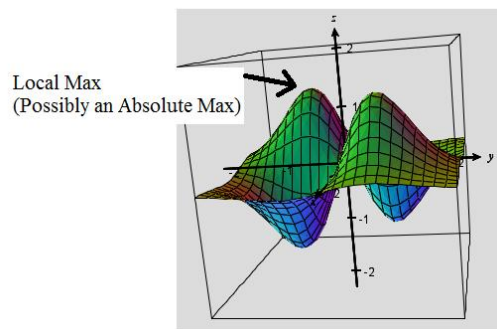
- A function f of two variables has a local maximum at (a,b) if $f(x,y) \leq f(a,b)$ when (x,y) is near (a,b) . The number $f(a,b)$ is called a local maximum value or simply a local maximum.
- A function f of two variables has a local minimum at (a,b) if $f(x,y) \geq f(a,b)$ when (x,y) is near (a,b) . The number $f(a,b)$ is called a local minimum value or simply a local minimum.

Note: We can make the notion of “near” precise by saying: “there exists some $\varepsilon > 0$ such that for every (x,y) in the disk centered at (a,b) of radius ε , $f(x,y) \leq f(a,b)$.”

Note: Local maxes/mins are z values!

Remark: If we replace the words “when (x,y) is near (a,b) ” in the above definitions with “for all (x,y) ” then we obtain the definition of absolute maximum and absolute minimum respectively, and we say f has an absolute maximum at (a,b) and f has an absolute minimum at (a,b) .

Ex: In the picture below a local max is identified. We cannot determine if it is a global max because it could be exceeded in a part of the graph that we cannot see.



Theorem: If a function of two variables f has a local maximum or minimum at (a,b) and the first-order partial derivatives of f exist there, then $f_x(a,b) = 0$ and $f_y(a,b) = 0$.

Proof: The proof is fairly straightforward; however, it is not included in the textbook.

Definition: A point (a,b) is called a critical point of $f(x,y)$ if $f_x(a,b) = 0$ and $f_y(a,b) = 0$ or if one of these partial derivatives does not exist.

Note: Critical points are where local minima/maxima MAY occur.

Example 1: Find all the critical points of the function $f(x, y) = x^2 + y^2 - 2x - 6y + 14$.

Theorem (Second Derivatives Test):

Suppose the second partial derivatives of f are continuous on a disk with center (a, b) , and suppose that

$f_x(a, b) = 0$ and $f_y(a, b) = 0$. Let $D = D(a, b) = f_{xx}(a, b)f_{yy}(a, b) - [f_{xy}(a, b)]^2$. Then,

1. If $D > 0$ and $f_{xx}(a, b) > 0$, then $f(a, b)$ is a local minimum.
2. If $D > 0$ and $f_{xx}(a, b) < 0$, then $f(a, b)$ is a local maximum.
3. If $D < 0$, then $f(a, b)$ is not a local maximum or local minimum. We say that (a, b) is a saddle point of f .
4. If $D = 0$, the test is inconclusive (i.e. it gives no information).

Proof: The proof is beyond the scope of this course. The key to proving this result is generalizing the notion of the second derivative test from calculus one.

Note: Results 1 and 2 are fairly easy to remember if one recalls the second derivative test from calculus 1.

Example 2: Find the local minimum and maximum value(s) and any saddle points of the function

$$f(x, y) = x^4 + y^4 - 4xy + 1.$$

Note: Language in the above example is very important. In particular, it is proper to say that there is a saddle point at $(0,0)$. It is also proper to say that there is a local minimum of -1 at $(1,1)$, or there is a local minimum of -1 at $(-1,-1)$. It would NOT be correct to say that there is a local minimum of $(1,1)$.

Example 3: Consider the function $f(x,y) = x^3 - y^3$. Show that $(0,0)$ is a critical point of f . Then, show that the second derivatives test is inconclusive in classifying $(0,0)$.

We now move to discussing absolute extrema.

Definition (intuitive): A closed set in $\mathbb{R}^2$ is one that contains all of its boundary points.

Note: For now, we will think of “boundary” in the intuitive sense. It is possible however, to give a mathematically rigorous definition of boundary point.

Ex: Based upon the above definition $\{(x, y) : x^2 + y^2 \leq 1\}$ is a closed set, and $\{(x, y) : x^2 + y^2 < 1\}$ is not a closed set.



Definition: A bounded set in $\mathbb{R}^2$ is one that is contained within some disk.

Theorem (Extreme Value Theorem for Functions of Two Variables):

If f is continuous on a closed, bounded set D in $\mathbb{R}^2$, then f attains an absolute maximum value $f(x_1, y_1)$ and an absolute minimum value $f(x_2, y_2)$ at some points (x_1, y_1) and (x_2, y_2) in D .

Proof: Beyond the scope of this course. This theorem is often proven in Real Analysis or General Topology.

Steps for Finding the Absolute Max or Min on a Closed Bounded Set:

1. Confirm that f is continuous.
2. Find the values of f at the critical points of f in D .
3. Find the extreme values of f on the boundary of D .
4. The largest of these values is the absolute maximum of f on D . The smallest of these values is the absolute minimum of f on D .

Example 4: Find the absolute maximum and minimum values of the function $f(x, y) = x^2 - 2xy + 2y$ on the rectangle $D = \{(x, y) : 0 \leq x \leq 3 \text{ and } 0 \leq y \leq 2\}$. Also, state where the absolute maximum and minimum values occur.

